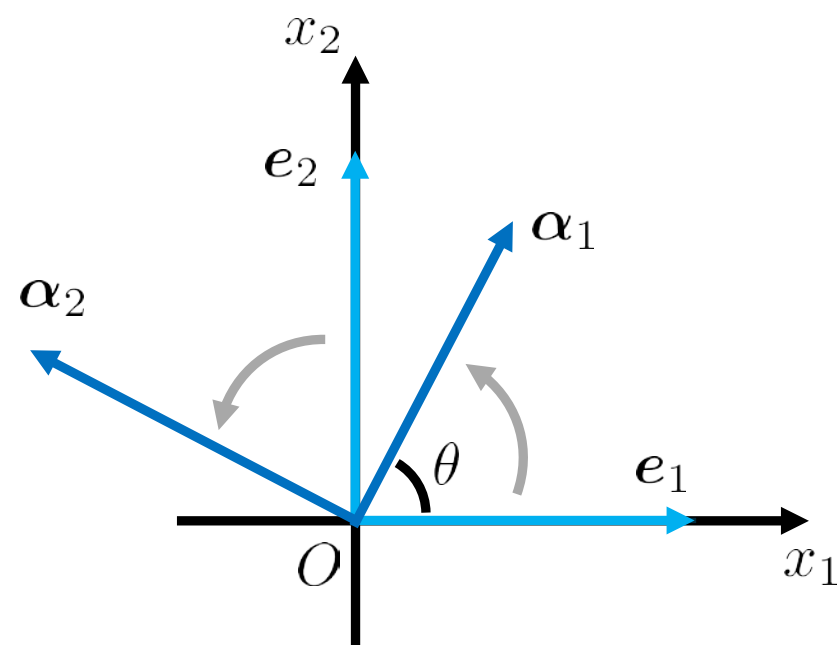
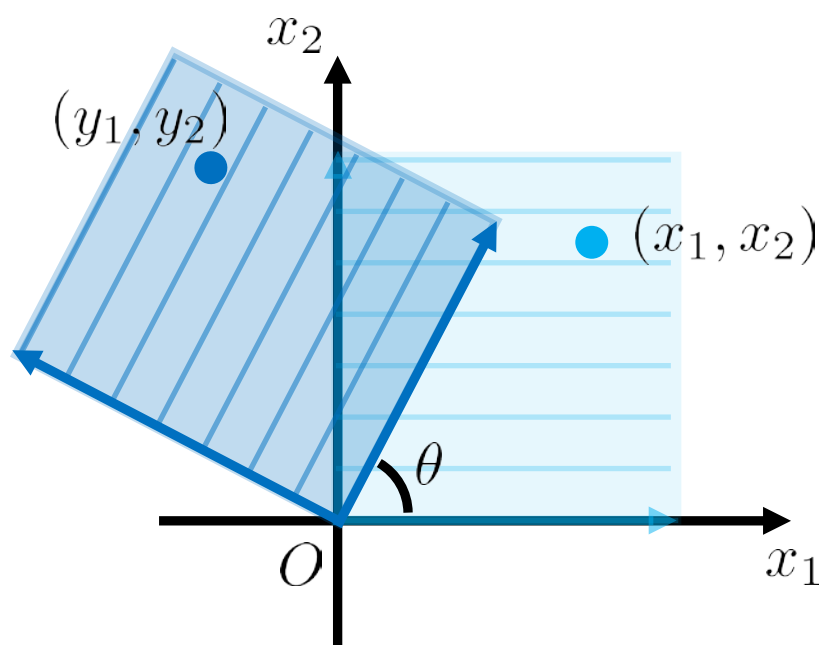




$$\begin{cases} y_1 = \cos \theta \cdot x_1 - \sin \theta \cdot x_2 \\ y_2 = \sin \theta \cdot x_1 + \cos \theta \cdot x_2 \end{cases} \iff \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\iff \mathbf{y} = A\mathbf{x} = (\boldsymbol{\alpha}_1 \ \boldsymbol{\alpha}_2)\mathbf{x}$$